

DecibelDefender: Using Virtual Reality for Promoting Sustainable Hearing Habits at Concerts and Festivals

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Figure 1: The concert scene of *DecibelDefender* VR experience

ABSTRACT

Noise-induced hearing loss is a growing concern among young adults attending concerts and festivals, where prolonged exposure to high-decibel environments can cause irreversible damage. *DecibelDefender* is a multiplayer VR experience designed to promote sustainable hearing habits by immersing users in a simulated concert environment. Participants experience progressive hearing loss and engage in interactive activities emphasizing the importance of hearing protection during concerts. Initial results shows an impact on behavioral intentions, with participants demonstrating greater preference for protective measures after trying the VR experience. This study highlights VR's potential to foster empathy and encourage preventive health behaviors.

Index Terms: Virtual Reality, Noise-Induced Hearing Loss, Preventive Health, Hearing Protection, Behavior Change.

1 INTRODUCTION

Noise-induced hearing loss (NIHL) is a growing health concern, particularly among young adults attending concerts and festivals. Prolonged exposure to high-decibel environments can lead to irreversible auditory damage, including hearing impairments and tinnitus [1]. Despite these risks, protective measures like earplugs are often underutilized due to misconceptions and social norms [4].

Virtual reality (VR) offers a unique medium for addressing public health challenges by creating immersive and engaging experiences. VR's ability to simulate real-world consequences in a controlled environment makes it particularly effective for raising awareness and fostering empathy [3]. Prior studies have shown its success in promoting behavior change across various health domains.

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Building on this potential, *DecibelDefender* is a multiplayer VR experience designed to promote sustainable hearing habits. The system immerses users in a dynamic concert environment, where they experience progressive hearing loss to simulate the consequences of unsafe listening behaviors. Through engaging gameplay and interactive narratives, users learn about preventive measures, such as wearing earplugs and taking breaks from loud environments, bridging the gap between awareness and behavior change.

2 METHODS

DecibelDefender was developed as a multiplayer VR experience using Unity 2021.3.1f1. The system consists of three main components: a tutorial, a concert simulation, and a social activity. These components aim to engage users while educating them about sustainable hearing habits.

The study was approved by the Aalborg University Ethics Committee (Ref: AAU084-1088362). All participants provided informed consent before participating in the experiment. A total of 30 participants (15 pairs) took part in the study. Data regarding the age of the participants was not collected. Participants were recruited at a social venue in Copenhagen (Absalon) targeting young adults attending concerts and music festivals.

2.1 VR Experience Design

The VR experience begins with a basic tutorial, followed by a concert simulation that immerses users in an environment where they experience progressive hearing loss, highlighting the consequences of unsafe listening habits. During this phase, the virtual guide, Magnus, provides narrative guidance on the importance of hearing protection. The experience concludes with a social activity featuring a game where participants throw virtual objects into cups, while Magnus explains the risks of not wearing earplugs during loud concerts.

2.2 Evaluation Protocol

Participants completed:

- **Knowledge Questionnaire:** Assessed participants' understanding of hearing health risks and preventive measures, measured as the number of correct answers.
- **Decisional Conflict Scale (DCS) [2]:** Evaluated confidence in decision-making related to hearing protection, including informed decision-making and clarity of values.

Statistical analyses included independent Shapiro-Wilk tests for normality on pre-test and post-test scores. A one-sample t-test on pre-post differences for the Knowledge Questionnaire and McNemar's Chi-square test for DCS data to analyze behavioral changes. Effect size was calculated using Cohen's w .

3 RESULTS

3.1 Knowledge Questionnaire

Descriptive statistics for the knowledge scores are shown in Table 1. Independent normality tests confirmed the pre-test ($p = 0.5$) and post-test ($p = 0.3$) distributions were normal. A one-sample t-test on pre-post differences revealed no significant improvement in knowledge scores ($p = 0.075$), though the mean post-test score was slightly higher.

	Before	After	Difference
Count	15	15	15
Mean	9.40	10.80	1.40
Standard Deviation	2.41	2.73	2.82
Min.	7.0	6.0	-4.0
Max.	15.0	14.0	7.0
Shapiro Wilk p-Value	0.0492	0.0668	0.478

Table 1: Descriptive statistics for Knowledge Questionnaire scores before and after the VR experience.

3.2 Decisional Conflict Scale (DCS)

The DCS results revealed significant behavioral changes in participants' confidence regarding hearing protection decisions. McNemar's Chi-square test confirmed a significant reduction in decisional conflict ($p < 0.001$), with a medium effect size (Cohen's $w = 0.35$). Figure 2 illustrates the shift in preferred protective measures.

3.3 Qualitative Feedback

Qualitative responses from dyadic interviews provided insights into user experiences. Common themes included: - **Engagement:** Users found the experience immersive and entertaining. - **Empathy:** Participants reported a deeper understanding of hearing impairment consequences. - **Challenges:** Some noted difficulty hearing instructions during the simulated hearing loss phase, and the term "beer pong" was seen as unscientific.

Figure 3 summarizes these themes.

4 DISCUSSION AND CONCLUSION

The results highlight the potential of *DecibelDefender* as an innovative tool for promoting sustainable hearing habits. Although the Knowledge Questionnaire showed no significant improvement, the DCS results and qualitative feedback demonstrated the system's ability to foster behavioral changes and increase awareness.

The lack of significant knowledge retention can be attributed to the immersive nature of the VR experience, which could have distracted users from educational content. Future iterations should incorporate clearer instructions and more structured knowledge delivery, such as interactive prompts or post-simulation reviews.

The DCS results underscore the system's impact on behavioral intentions. By simulating the consequences of unsafe listening

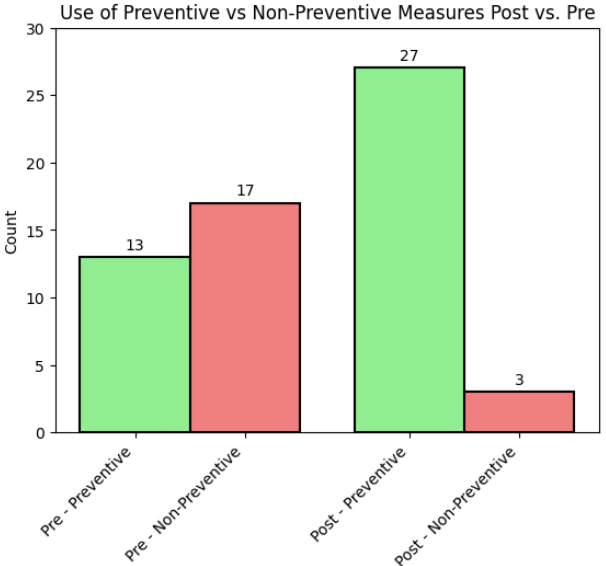


Figure 2: Bar chart displaying participants' preferred preventive measures before and after the VR experience.

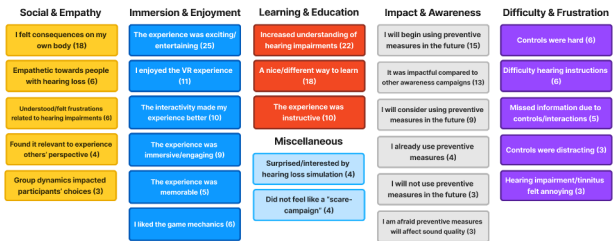


Figure 3: Affinity diagram summarizing qualitative feedback categories.

habits, *DecibelDefender* effectively engaged participants and encouraged protective measures. The medium effect size suggests meaningful behavioral shifts, emphasizing VR's potential as an empathy-driven educational tool.

Despite positive feedback, usability issues—such as difficulty with controls and hearing instructions—highlight areas for refinement. Replacing informal terms like "beer pong" with descriptive terminology and providing more context for the interactive auditory game will further enhance the intervention's scientific rigor.

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